

An anatomical illustration of the human brain and cranial nerves, rendered in a dark, textured style. The brain is shown from a superior view, with the cerebellum and brainstem visible at the bottom. Various cranial nerves are labeled with numbers 1 through 9. The text 'Brain & Cranial Nerves II' is overlaid in a large, white, serif font.

# Brain & Cranial Nerves II

BIOL2251

Chapter 14

Section 14.9 – 14.10

# Lecture Quiz



**Test what you know!**



**9 questions** (MC + **Matching**)

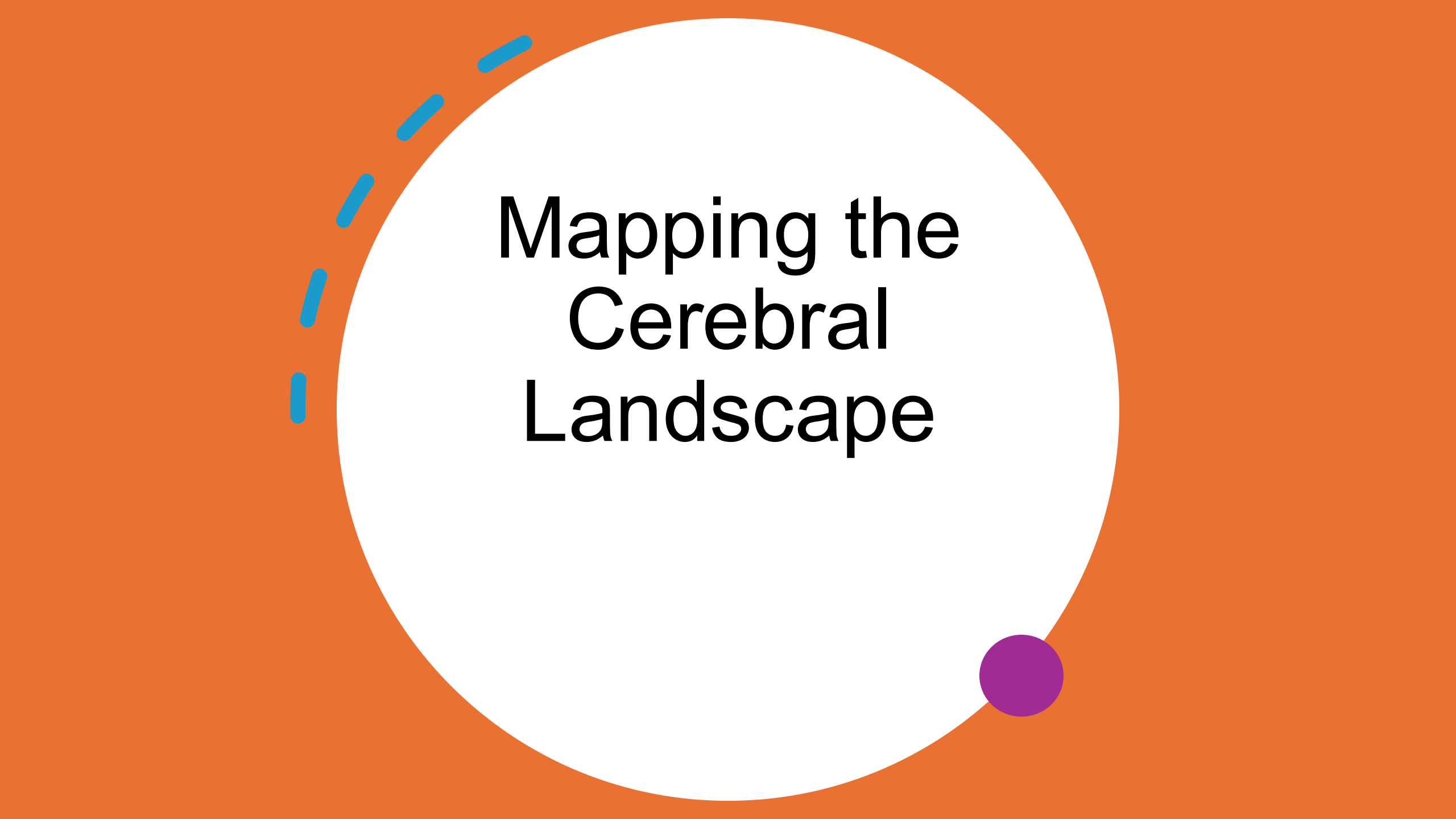


**Quiz automatically submits at 2:30 PM**



**Password = cortex**

**Make sure you sign in before the end of lecture!**



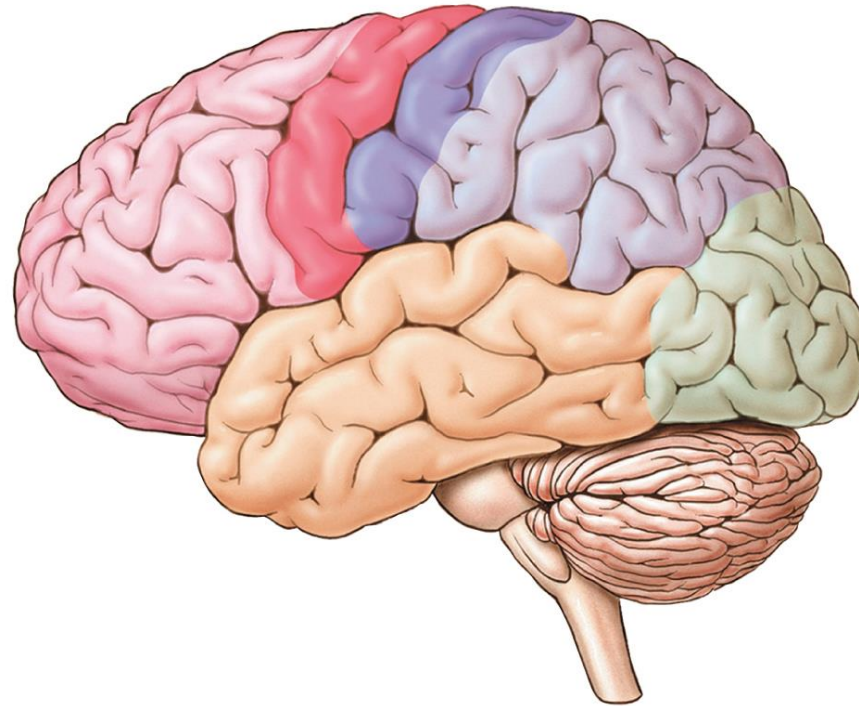
# Mapping the Cerebral Landscape

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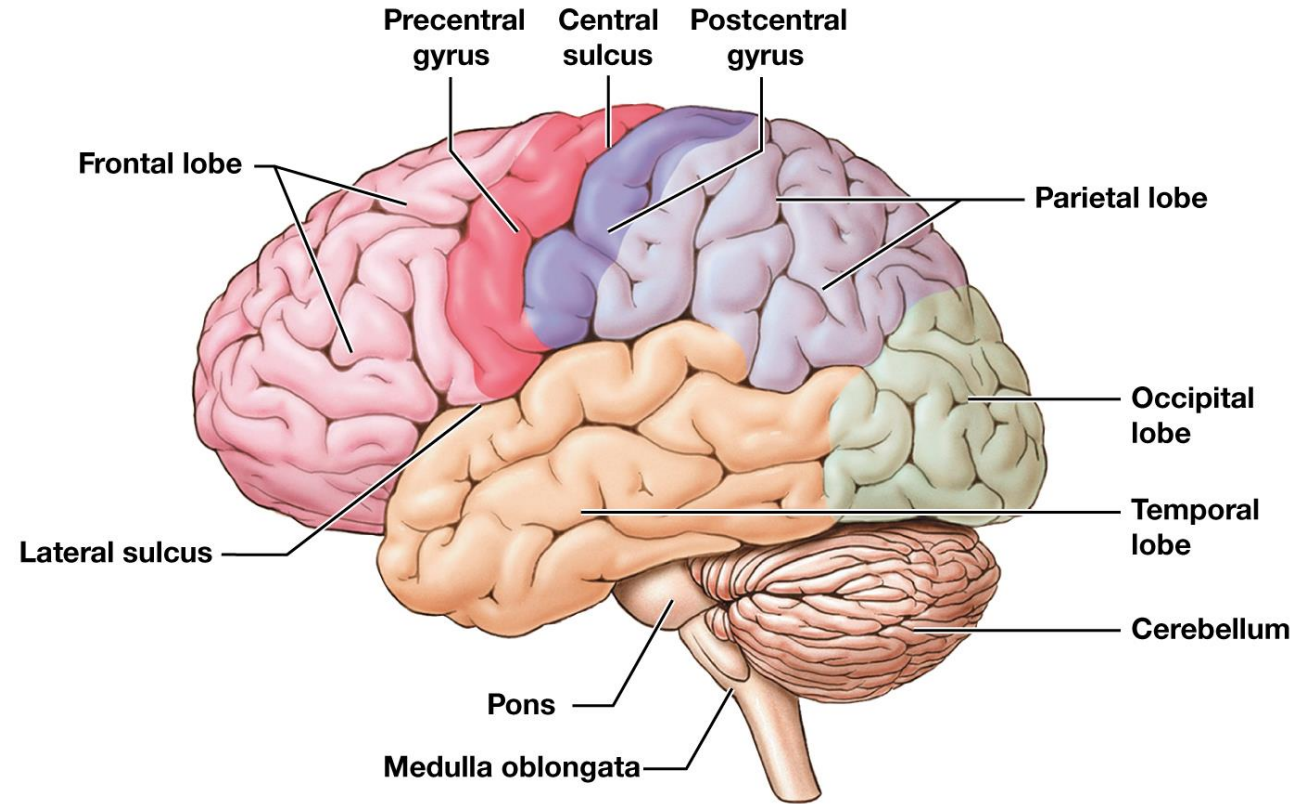
- **Instructions**

- I will project an unlabeled brain from the lateral view
- **Working in pairs, I want you to identify**
  - **4 lobes**
  - **2 gyri**
  - **2 sulci**
- We'll then check the major lobes, gyri, and sulci
- **Bloom's taxonomy = Remember/Understand**

# Mapping the Cerebral Landscape



# Mapping the Cerebral Landscape



**b** Lateral view



# White Matter Communication

# White Matter Communication

- **Instructions**

- Work and discuss in small groups
- Based on the description, **classify the white matter fibers as association, commissural, or projection**

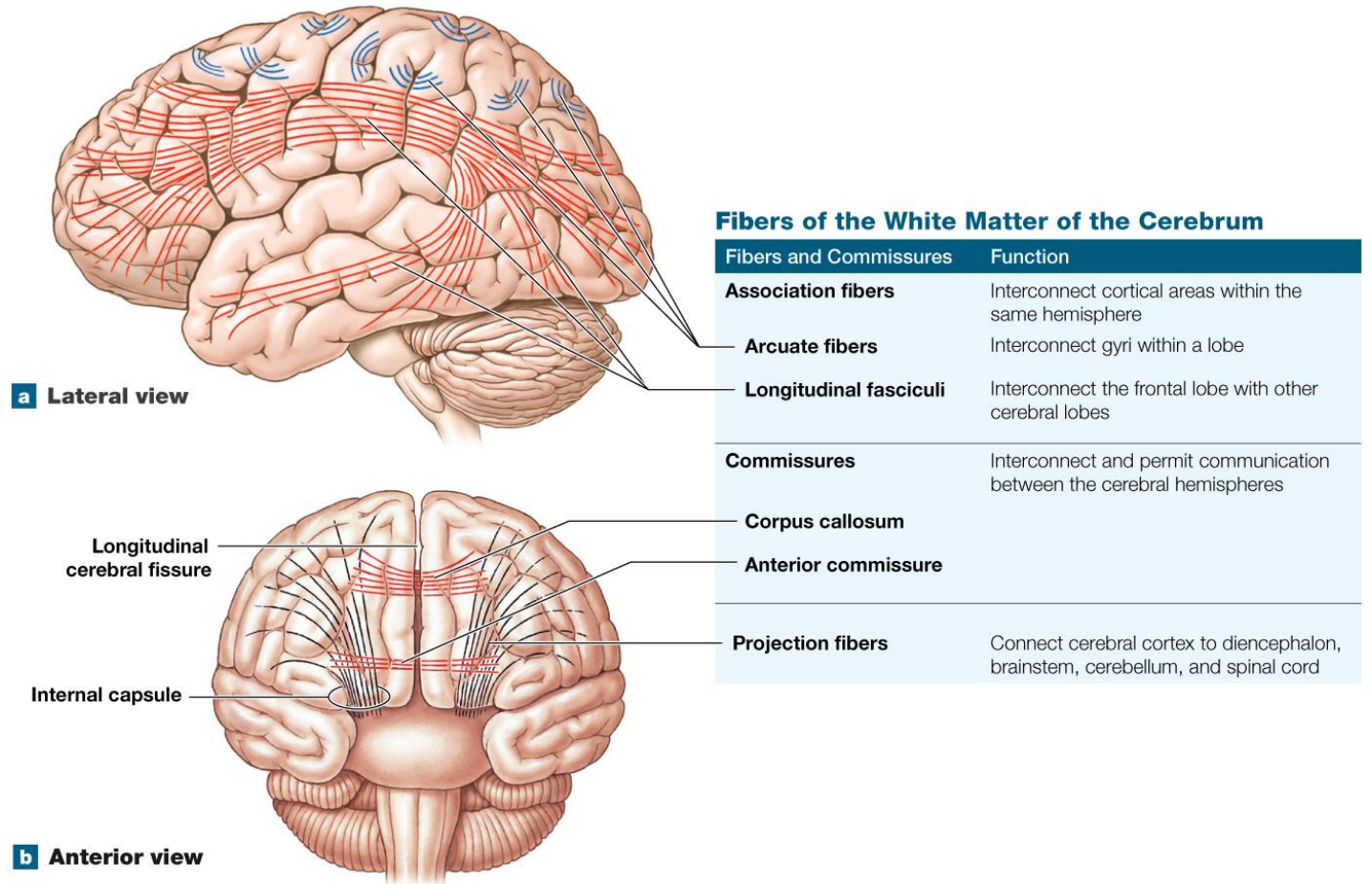
- **Bloom's taxonomy = Understand/Apply**

| Scenario                                                                                | Fiber Type? |
|-----------------------------------------------------------------------------------------|-------------|
| Neurons connecting the prefrontal cortex to the premotor cortex in the same hemisphere. |             |
| Axons carrying motor commands from the cortex to the brainstem.                         |             |
| Axons linking the left and right hemispheres for communication.                         |             |

# White Matter Communication

| Scenario                                                                                | Fiber Type |
|-----------------------------------------------------------------------------------------|------------|
| Neurons connecting the prefrontal cortex to the premotor cortex in the same hemisphere. | A          |
| Axons carrying motor commands from the cortex to the brainstem.                         | P          |
| Axons linking the left and right hemispheres for communication.                         | C          |

# White Matter Communication





# Brain Area Sorting

# Brain Area Sorting

- **Instructions**

- Work and discuss in small groups
  - **Match each of the brain areas to its correct description/function**
- **Bloom's taxonomy = Apply/Analyze**

# Brain Area Sorting

| Brain Area                     | Function (Shuffled)                                                 |
|--------------------------------|---------------------------------------------------------------------|
| Primary motor cortex           | A. Interprets auditory information like words and tones.            |
| Primary somatosensory cortex   | B. Receives general sensory input (touch, pressure).                |
| Primary visual cortex          | C. Initiates voluntary muscle movement.                             |
| Primary auditory cortex        | D. Receives sound information.                                      |
| Olfactory cortex               | E. Receives visual information.                                     |
| Gustatory cortex               | F. Processes taste sensations from the tongue.                      |
| Premotor cortex                | G. Coordinates learned movements.                                   |
| Visual association area        | H. Receives smell information from nasal receptors.                 |
| Auditory association area      | I. Interpretation and cognition of sensory input (touch, pressure). |
| Somatosensory association area | J. Interprets visual information.                                   |

# Brain Area Sorting Key

| Brain Area                     | Function                                                            | Lobe      |
|--------------------------------|---------------------------------------------------------------------|-----------|
| Primary motor cortex           | C. Initiates voluntary muscle movement.                             | Frontal   |
| Primary somatosensory cortex   | B. Receives general sensory input (touch, pressure, pain).          | Parietal  |
| Primary visual cortex          | E. Receives visual information                                      | Occipital |
| Primary auditory cortex        | D. Receives sound information.                                      | Temporal  |
| Olfactory cortex               | H. Receives smell information from nasal receptors.                 | Temporal  |
| Gustatory cortex               | F. Processes taste sensations from the tongue.                      | Frontal   |
| Premotor cortex                | G. Coordinates learned movements.                                   | Frontal   |
| Visual association area        | J. Interprets visual information (e.g., letters, words)             | Occipital |
| Auditory association area      | A. Interprets auditory information like words and tones.            | Temporal  |
| Somatosensory association area | I. Interpretation and cognition of sensory input (touch, pressure). | Parietal  |



# Integrative Areas and Lateralization

# Integrative Areas and Lateralization

- **Instructions**

- Work in pairs or small groups
- You will be given a short case where **you will identify which area and side of the brain has been affected**
  - **Take 2 minutes to think and discuss as a group**
  - **Write your answer on a notecard and turn it in**
- We will review each case

- **Bloom's taxonomy = Apply/Analyze**

# Integrative Areas and Lateralization

1. A patient speaks fluently but makes no sense and cannot understand language.

- **Wernicke's area (left hemisphere)**

2. A patient understands language but can only speak in short, broken phrases.

- **Broca's area (left hemisphere)**

# Integrative Areas and Lateralization

3. A patient with head trauma is intelligent but now acts impulsively, shows poor judgment, and seems indifferent to consequences.

- **Prefrontal cortex (bilateral frontal lobes)**

4. A patient can sing lyrics of a song but cannot speak those same words normally.

- **Left hemisphere (language) with compensation by right hemisphere (music/melody)**

# Brain Activity and EEG Analysis



# Brain Activity and EEG Analysis

- **Instructions**

- Work in pairs
- I will give a short scenario and **you will identify which type of brain waves best fits the situation**
  - **Alpha, beta, theta, delta**
- All pairs will 'vote' by raising their hand as I call out the type of brain waves

- **Bloom's taxonomy = Understand/Apply**

# Brain Activity and EEG Analysis

- A relaxed student sits quietly with eyes closed, breathing slowly before an exam → **Alpha**
- A student is alert and focused, rapidly solving complex anatomy quiz questions → **Beta**
- A student is stuck in absurd traffic on their commute home after classes and work → **Theta**
- A person in deep, restful sleep shows slow, large amplitude brain waves on EEG → **Delta**



# Cranial Nerves

# Cranial Nerves

- **Instructions**

- Work in pairs
- I will read a **short clue about a cranial nerve's primary function**
- Quickly discuss and **identify the name of the cranial nerve**
- First pair with their hand up = first attempt at the answer
- Follow-up to **identify if the nerve is sensory, motor, or mixed**
- **Bloom's taxonomy = Remember/Apply/Analyze**

| #    | Nerve             | Clue                                                                                                 | Function Summary                            | Type    |
|------|-------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------|---------|
| I    | Olfactory         | "This nerve carries the sense of smell directly to the brain."                                       | Smell                                       | Sensory |
| II   | Optic             | "This nerve transmits visual information from the retina to the brain."                              | Vision                                      | Sensory |
| III  | Oculomotor        | "This nerve moves most of the eye muscles and constricts the pupil."                                 | Eye movement; pupil constriction            | Motor   |
| IV   | Trochlear         | "This nerve moves the superior oblique muscle, allowing you to look down and to the side."           | Eye movement                                | Motor   |
| V    | Trigeminal        | "This large nerve carries facial sensation and controls the muscles for chewing."                    | Facial sensation; chewing                   | Mixed   |
| VI   | Abducens          | "This small nerve moves the lateral rectus muscle to abduct the eye."                                | Eye movement                                | Motor   |
| VII  | Facial            | "This nerve controls facial expressions and carries taste from the front of the tongue."             | Facial muscles; taste (anterior 2/3 tongue) | Mixed   |
| VIII | Vestibulocochlear | "This nerve helps you hear and maintain balance."                                                    | Hearing; balance                            | Sensory |
| IX   | Glossopharyngeal  | "This nerve controls swallowing and carries taste from the back of the tongue."                      | Taste (posterior 1/3); swallowing           | Mixed   |
| X    | Vagus             | "This nerve extends into the thoracic and abdominal cavities, controlling heart rate and digestion." | Autonomic control of viscera; speech        | Mixed   |
| XI   | Accessory         | "This nerve allows you to shrug your shoulders and turn your head."                                  | Neck and shoulder movement                  | Motor   |
| XII  | Hypoglossal       | "This nerve controls tongue movement for speech and swallowing."                                     | Tongue movement                             | Motor   |